

EXHAUSTIVE INFERENCES IN REPLIES

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1 Introduction

Used as replies¹ to constituent questions, utterances often yield exhaustive inferences.

- (1) Among the musketeers, who is inside the barracks?
Porthos and Aramis are inside.
→ *Athos isn't inside.*

Groenendijk and Stokhof (1984) proposed that these exhaustive inferences are the product of a semantic operation EXH². In its modern formulation, EXH takes a proposition p and a set of contextually provided alternatives Alt and delivers a proposition representing the exhaustive reading of the answer. For Groenendijk and Stokhof (1984), the operation EXH is necessarily triggered in a question environment and the meaning of the question asked is a direct argument of EXH ; in modern terms, we could say that the question serves as the argument Alt . Later, in Chierchia, Fox, and Spector (2012), EXH acquires an independent life outside of question/answer pairs, being also responsible for the generation of scalar inferences. Alt becomes a free parameter to be set by context. In this view, some pragmatic principle must come and regulate the selection of Alt . Whatever the principles are, they must explain the reliable appearance of exhaustive inferences in replies.

Through a series of progressive refinements, this piece seeks to spell out the theory of the pragmatics of EXH, how the argument Alt is selected, focusing on question/answer pairs, and

¹I use the term ‘reply’ rather than ‘answer’, because ‘answer’ is sometimes used to mean replies that match the template of the question (either term answers or full sentential answers). As the sequel will hopefully show, there are interesting facts about exhaustivity inferences when the reply does not match the template of its answer.

²They argue against the Gricean pragmatic account of the earlier Groenendijk and Stokhof (1984).

how this selection connects to both the form of the question and the form of the reply. Many of the conclusions reached here are similar to ones drawn in disseminated corners of previous literature. As these conclusions are compiled however, new and unnoticed challenges will arise that will be noted and left open here.

1 Hypothesis 1: Exhaustivity inferences only depend on the question's form

If questions denote sets of propositions, it would stand to reason that these propositions provide the set of alternatives for EXH, as in [Groenendijk and Stokhof \(1984\)](#). The pragmatic principle in (2) formalizes this:

- (2) In a sequence 'Q? A', A is interpreted as $\text{Exh}([A], [Q])$

This view is defeated by the fact that semantically equivalent forms do not yield the same exhaustification inferences when uttered against the same question. In (3), the French adjective *protéiné*³ 'proteinated' literally means the same as *contient des protéines* 'contains proteins' but does not convey the exhaustivity inference that the latter expression does.

- (3) Qu'est-ce que ça contient ?
 What-Q it contains
 a. Ca contient des protéines.
 It contains some proteins.
 → *it's made of proteins and nothing else (among the relevant chemical compounds)*
 b. C'est protéiné.
 It's 'proteinated'
 → *it's made of proteins and nothing else (among the relevant chemical compounds)*

Likewise, in English, all forms in (4)a, (4)b and (4)c have the same underlying literal meaning prior to exhaustification, but differ in the exhaustivity inferences they give rise to.

- (4) How many living parents does this lamb have?
 a. It has one. → just one
 b. It has at least one. → /just one
 c. It isn't an orphan. → /just one

2 Hypothesis 2: exhaustivity inferences is only determined on the basis of the answer's form

The difference between *c'est protéiné* 'it's proteinated' and *ça contient des protéines* 'it contains proteins' is also observed with *only* outside the context of questions. Something in the former expression resists exhaustification, in a way that doesn't apply to the latter expression.

- (5)

³The French word is natural, unlike the English neologism 'proteinated', hence the resort to French.

- a. ??C'est seulement protéiné_F.
 It's only proteinated.
 Intended: *it's only made of proteins*
- b. Ca contient seulement [des protéines]_F.
 It's contains only proteins_F.
 Intended: *it's only made of proteins*

If the contrast in (5) exist whether or not these sentences are used as answers to a question asked, it suggests that a question asked cannot give a reply new alternatives that it could not have in isolation. This leads us to the following view on the interpretation of answers:

- (6) In a sequence 'Q? A', A is interpreted as $\text{EXH}(\llbracket A \rrbracket, \text{ALT}_A)$ where ALT_A is a set of alternative propositions determined from the syntactic form of A, in particular focus marking.

As stated, (6) denies that a question can have any influence on the interpretation of its replies. But this is patently false: as (7) attests, the same reply, accented in the same manner, uttered against different questions convey different exhaustivity inferences (a similar argument is found in [Groenendijk and Stokhof \(1984\)](#)).

- (7) Context-dependence of EXH
- a. Which MIT professors will find this interesting?
 Danny_F will. $\leadsto$ *no other MIT professor will find this interesting.*
- b. Which amateur basketball players will find this interesting?
 Danny_F will. $\leadsto$ *no other amateur basketball player will find this interesting.*

3 Hypothesis 3: relevance-based domain selection

While questions may not add alternatives that a reply does not have, they may remove from considerations alternatives that a reply can have. There, we reach a somewhat common view:

- (8) In a sequence 'Q? A', A is interpreted as $\text{EXH}(\llbracket A \rrbracket, C)$ where:
- $C \subset \text{ALT}_A$
 - ALT_A is a set of propositions determined from the syntactic form of A, in particular focus marking.

(8) preserves the difference between *protéiné* 'proteinated' and *contient des protéines* 'contains proteins': the former still does not have the alternatives that it needs to be interpreted exhaustively. But it allows the observed variation in (9) and (10).

- (9) Which MIT professor will find this interesting?
 Danny_F will. $C = \{x \text{ will find this interesting} \mid x \text{ is an MIT professor} \}$
- (10) Which amateur basketball player will find this interesting?
 Danny_F will. $C = \{x \text{ will find this interesting} \mid x \text{ is an amateur basketball player} \}$

But how is C selected? The question could be settled by appeal to general principles of pragmatics ; the domain C is selected in much the same way that covert domains of quantifier are disambiguated: in appropriate contexts, (11) may mean any of (11)a to (11)d. Complex pragmatic reasons explain why one interpretation stands out on a given occasion.

- (11) Everyone discusses the theory.
 a. = *everyone on Earth*,
 b. = *everyone studying phonology at MIT*,
 c. = *everyone at Harvard*
 d. = *every linguist at MIT*

While covert domain selection in general is a complex problem, simple factors can be identified in question/answer contexts: relevance and the need to provide complete answers. Thus, *everyone* in (12) is naturally interpreted as (11)d, and not as (11)a, (11)b or (11)c, presumably because it is the only interpretation that provides a complete answer to the question.

- (12) _ Among the linguists currently at MIT, who discusses this theory?
 _ Everyone.

Adapting the observation in (12) to the domain of the exhaustification operator, we may supplement (8) with a pragmatic principle like (13)

- (13) To the extent possible, pick C so that EXH([A], C) is a complete answer to the question asked.

Note that (13) requires “a complete answer” (in other words, the proposition must denote a cell of the question’s partition), rather than “an answer”. This is necessary if we want to avoid ‘under-exhaustification’, i.e. interpreting (14) as exhaustive with respect to a smaller set of alternatives

- (14) Which MIT professor will find this interesting?
 Danny_F will. → *no other syntax/semantics professor*
 C = {*x* will find this interesting | *x* is an MIT syntax/semantics professor }

The principle in (13) should, in the final instance, derive from Gricean maxims like (15). But we leave it at that at this stage.

- (15) Maxim of Quantity: make your contribution as informative as possible
 Maxim of Relation: make your contribution relevant

Before moving to further complications, a remark is in order. In previous literature, different principles for the selection of C have been proposed: in Magri (2011), C must be the subset of ALT_A containing only relevant propositions; in Bar-Lev (2021); Bar-Lev (2024), C must be picked so that EXH([A], C) is relevant and the closest to ALT_A it can be. In both of these works, C can be such that EXH([A], C) is a contextual contradiction. The null hypothesis presented here, by contrast, is that the domain of EXH is set like the domain of any other operator (e.g. *everyone*). This hypothesis likely comes into conflict with previous works: the domain of operators like *everyone* is never set in such a way that it would generate contextual contradictions and so we do not expect that domain selection for EXH would too. A dilemma arises (cf in this connection Magri (2011) fn. 5): either we accept EXH-specific rules of domain selection and must try to explain why such rules should exist or we risk losing the empirical purchase that these EXH-specific principles afford.

4 Hypothesis 4: question shift

There are two opposite issues with our current working hypothesis. First, exhaustive inferences are observed in replies even when these inferences don't make a complete relevant answer. For instance, one infers from (16)b that the president hasn't assured the press that a decision on budget has been taken.

(16)

- a._ Has the government decided on a budget proposal?
- b._ The prime minister and the secretary of state have assured the press that it has.

Likewise, (17)b leads to exhaustive inferences that Jane didn't sell cocaine to all senators, that she didn't counterfeit art, etc.

(17)

- a._ What type of criminal is Jane?
- b._ She sold cocaine to some of our senators.

The principle of domain selection stated in (13) of the previous section doesn't technically make mispredictions here. It makes no predictions, since neither (16)b nor (17)b can be exhaustified in a way that provides a complete answer to the question asked. So, at first blush, it may be dismissed as a question for future research. However, the examples in (16) and (17) have a natural connection to the challenge we now turn to.

The second (and opposite) challenge genuinely threatens (13) as stated. Consider the contrast between (18)a and (18)b.

(18) (Meeting with tax attorney:) How many dependents did you claim last year?

- a.# I claimed a dependent.
- b.I claimed one dependent.

The infelicity of (18)a is surprising, given our theory so far. Indeed, the utterance in (18)a, in isolation, can convey that I have one and only one dependent, thereby providing a complete answer to the question asked. Some application of EXH is therefore capable of making (18)a into a complete answer to the question in (18). The principle of domain selection in (13) cannot find any reason why (18)a could not serve as an answer.

(19) I have a dependent.

→ *I don't have several dependents.*

This second challenge suggests that the meaning of an utterance alone (including the contributions of EXH) does not determine whether it can be a good answer to a question. Going one step further, with the support of previous literature (Rooth 1992; Roberts 1996), we may say that an utterance carries in its form indications of what question it is aimed to answer. The problem with '*I have a dependent*' in this view is not its meaning, but that it isn't answering the right question. More precisely, it is thought that the question evoked by an utterance is given in part by its focus structure. Namely, the question evoked is a partition formed from a subset of

those alternatives obtained by replacing the focus-marked constituent with *appropriate* alternatives.

‘*I have a dependent*’ may be focus-marked in different ways but no choice of focus-marking, I claim, indicates an answer to the question that was asked. (20)b, for instance, may answer the question ‘what do you have?’, (20)c the question ‘what’s the world like?’ or perhaps ‘do you have a dependent?’.

- (20) I have a dependent
 - a. I have A_F dependent.
 - b. I have [a dePENdent]_F.
 - c. [I have a dePENdent]_F.

(20)a deserves more discussion. If ‘appropriate alternatives’ to the indefinite *a* contain the numerals *two* or *three*, then (20)a could serve as an answer to the ‘how many?’ question. So numerals probably do not count as alternatives to the indefinite *a*. Potentially, an independent line of evidence could come from the study of contrastive focus and examples like (21)a and (21)b.

- (21)
 - a. # I have *a* dependent, not *two* dependents.
 - b. # I have *a* dependent, not *three* dependents.
 - c. I have *a* dependent, not *several* dependents.
 - d. I have *an* answer, not *the* answer.

More would need to be said as to why this is so. Regardless, the idea that utterances may intrinsically carry signs of what questions they answer helps with the first challenge in (16) and (17) too. If utterances (loosely) signal which question they are answers to, it seems reasonable to think that the exhaustive inferences they give rise to are computed taking into consideration the question signaled, rather than the question asked. (22) and (23) illustrate how this could go. In essence, we distinguish the Question Under Discussion (QUD) from the question asked. The domain must be picked in such a way that the utterance is a complete answer to the QUD, but not necessarily the question asked.

(22) To the extent possible, pick *C* so that $EXH(\llbracket A \rrbracket, C)$ is a complete answer to the QUD.

(23) The QUD is a subset of the focus alternatives of *A*.

So, for instance, in (24), the focus marking suggests a QUD of the form ‘*which government official said what?*’. With the appropriate domain, this leads (at least) to the inference that the president didn’t make any assurances.

- (24) _ Has the government decided on a budget proposal?
 _ [The prime minister and the secretary of state]_F have assured [the press that it HAS]_F.
 (Q = who has said what?)

A remark before moving on: focus marking seems to play two roles here. It signals the QUD but also constrains the alternatives to EXH. This dual role of focus marking is recognized in [Katzir](#)

(2024). Yet, the two functions of focus marking appear connected. An important project for future research would be to draw out and explain this connection.

5 Shift appropriateness

Divorcing QUDs from the question asked brings us dangerously close to the view dismissed in section 2, that the exhaustive reading of a reply does not depend at all on the question asked but solely on the form and meaning of the answer. The case was dismissed by examples like (25). But a similar problem rears its head again if we allow the QUD in (25)a to be, in spite of the question asked, ‘which amateur basketball player will find this interesting?’ or vice-versa in (25)b. Somewhere in the theory of pragmatics, there must therefore be constraints on how the QUD relate to the question asked.

(25)

a. Which MIT professor will find this interesting?

DA_{nny}_F will. $\rightsquigarrow$ *no other MIT professor will find this interesting.*

b. Which amateur basketball player will find this interesting?

DA_{nny}_F will. $\rightsquigarrow$ *no other amateur basketball player will find this interesting.*

The proposition here is simple: in essence, a question asked is a proposal to set the QUD to a certain partition. The proposal may be accepted or rejected. Rejections may take the form of an utterance that signals a QUD different from the question asked. In *cooperative* communication, rejections of a proposed question could occur when the answerer has a reason to think that a better question could be answered. What counts as better depends on the shared purpose of the inquirer and the answerer (as understood by the answerer). (26) is an example of this type of dynamics. While the inquirer asks about who all is coming to the party, the answerer, through focus marking, signals an answer to ‘is Jane coming?’. The subtext is that, according to the answerer, the inquirer is in fact more interested in Jane’s coming to the party than any other invitee (perhaps Jane is a romantic interest of the inquirer?).

(26) _ Who is coming to the party?

_ [Jane is COming]_F, my friend.

Likewise, the waiter in (27) judges that ‘*where are the restrooms?*’ is a better question to answer than the question asked, because they naturally infer, from the very question asked, that the inquirer’s goal is to find restrooms.

(27) _ Does your restaurant have restrooms?

_ They’re [in the BACK]_F

Shifts from the question asked, in cooperative conversations, are therefore not gratuitous nor automatic. First, the new QUD must be a better way for the speaker to address the goals of the inquirer, as understood by the answerer. Second, it must also be clear to the inquirer what the new QUD is and why the answerer judged it a better option. For instance, the sequence (28) will sound like an abrupt change of topic, unless the inquirer can infer that there is a principled connection between Danny writing the hand-out and there being typos.

(28) Does this hand-out have typos?

DA_{nny}_F wrote it.

(29) In cooperative conversation, a QUD may differ from the question asked if:

- answering the QUD better fulfills the goals of the conversation
- the inquirer will recognize that this is so

6 Polarity shift

Our matured pragmatic theory faces a final hurdle, which I will only solve speculatively. Spector (2006) points the following contrast: when the reply to a question contains information about the anti-extension⁴ of a question predicate, no exhaustivity inferences is felt.

(30) Among the musketeers, who is inside the barracks?

- | | |
|------------------------|--|
| a. Porthos is outside. | →/Aramis isn't inside. →/Aramis isn't outside. |
| b. Porthos is inside. | → Aramis isn't inside |

Spector's solution requires admitting that the question asked may set the alternatives for EXH. We dismissed this option in section 2, because it ran the risk of predicting exhaustive inferences with forms that were semantically equivalent to an answer. In the spirit of the proposal here, a different explanation may be sketched. With accenting as in (31), no exhaustivity inference is indeed predicted. The QUD signaled is presumably the polar question 'is Porthos outside?'. A shift to such a QUD might indeed a felicitous discourse move if there is common understanding that the fact that Porthos is outside is particularly noteworthy.

(31) [Porthos is OUTside]_F

The real question is why the focus-marking as in (32) is not available. With this marking, EXH may deliver a complete answer to the question asked.

(32) PORthos_F is outside.
→ *who is outside?*

It may be that this accent signals a QUD '*who is outside?*' different from the question asked. Absent any reason to shift from '*who is inside?*' to '*who is outside?*' (that can be recognized by the inquirer), the accent in (32) would appear odd. But there is a flaw in this reasoning: QUDs are partitions of the common ground and thus '*who is outside?*', the proposed QUD, is exactly equal to the question asked. (32) thus does not signal a question shift and the challenge remains. Speculatively then and out of despair perhaps, I want to suggest that QUDs themselves are not partitions of the common ground. Examples like (33) have been used to show that interrogative sentences (the linguistic forms) do not denote partitions at the semantic level.

(33)

- | |
|--|
| a. Jane was surprised at who was inside. |
| b. Jane was surprised at who was inside. |

⁴ Spector's original examples involve negations like 'Not Aramis'. But these examples have a conceivable explanation in terms of ALT_A and the semantics of EXH.

There hasn't been, to my knowledge, any claim that unembedded questions like 'who is inside?' and 'who is outside?' make different contributions at the pragmatic level. Both, it is assumed, give rise to one and the same partition (the QUD for subsequent exchange). But evidence to the contrary might come from (34). In this example, I submit there is a preference to ask the question in one polarity versus another, depending on the sample that must be acted on.

- (34) Context: Some of the Petri dishes on the table have been contaminated.
- a. Context 1: safe samples must be left on the table, contaminated samples go to the incinerator.
 - b. Context 2: safe samples must be put in the fridge, contaminated samples must be left on the table.
 - c. Which samples are contaminated? *Preferred in context 1*
 - d. Which samples are safe? *Preferred in context 2*

These marginal polarity preference could be interpreted as revealing that, in asking 'which samples are safe?', the inquirer signals a preference to know, of the true answers, that they are true over knowing, of the false answers, that they are false (though they may desire to know both). This minute difference between questions of different polarities might explain why the shift from 'who is inside?' to 'who is outside?' forced by (32) is a genuinely significant. This shift is not gratuitous and must meet the constraints laid out in (29). Short of any perceivable justification for this shift, the reply in (32) could sound inadmissible. An untested prediction is that the accenting in (32) might be acceptable if the context is set up so that the speaker's signaling that 'who is outside?' is preferable to 'who is inside?' could be justifiable.

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